Neural Network-Based Estimation of Time-Varying Frequencies for Sinusoidal Signals

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This study investigates neural network-based estimation of time-varying frequencies using three architectures: an original multilayer perceptron (MLP), a modified MLP (mMLP), and a one-dimensional convolutional neural network (1D CNN). The original MLP, developed for constant-frequency estimation, showed limited performance on time-varying signals. Increasing the network capacity significantly improved estimation accuracy, with the mMLP achieving a test root mean square error (RMSE) of 20.823 Hz. The 1D CNN achieved the best performance, reducing the RMSE to 14.845 Hz, approximately 29% lower than that of the mMLP. Despite its higher accuracy and parameter efficiency, the CNN required greater computational time. Furthermore, its performance deteriorated when applied to real experimental vibration signals, highlighting the gap between synthetic and real-world data. These findings demonstrate the potential of neural networks for time-varying frequency estimation while emphasizing the need for improved robustness under realistic measurement conditions.

I. Introduction

Accurate frequency estimation is important in various engineering fields. In mechanical systems, frequency information is instrumental in identifying structural characteristics, monitoring system variations, and facilitating active vibration and noise control [1, 2]. Moreover, in aerospace applications, such as spacecraft and launch vehicles, accurate frequency estimation is vital for detecting vibrations induced by flexible structures. This capability is essential for preventing resonance and enabling stable attitude control [3, 4].

Several methods have been developed for frequency estimation. One of the simplest methods is the zero-crossing technique, which offers simplicity and ease of implementation [5]. In the frequency-domain approach, the Fourier Transform is a commonly employed method for frequency estimation [6]. However, this method usually relies on batch processing, making it difficult to apply to real-time or time-varying signals. In contrast, various time-domain techniques have also been explored, such as recursive least squares (RLS) [7], adaptive notch filter (ANF) [3, 4], and Kalman filter-based approaches [8–10]. The Kalman filter is among the most widely studied algorithms for multi-component, noisy, or time-varying signals, due to its model-based ability to handle model errors and measurement noise. On the other hand, neural networks offer a data-driven approach that can learn relationships between measured signals and their corresponding frequencies without requiring an explicit structural dynamics model.

Recent advances in machine learning have motivated the application of neural networks to aerospace engineering problems. In particular, learning-based frequency estimation could provide a more convenient, generalizable, and potentially robust alternative to conventional approaches when vibration characteristics are uncertain. Previous research demonstrated that a relatively simple feedforward neural network could estimation the frequency of noisy sinusoidal signals under constant-frequency conditions [11]. However, real signals such as spacecraft structural vibration may vary over time due to changes in operating conditions, system dynamics, and structural changes. Therefore, the performance of such a neural network under time-varying frequency conditions requires further investigation.

The objective of this study is to develop neural network-based estimation of time-varying frequencies using synthetic noisy sinusoidal signals. Three neural network architectures are considered: (1) the original multi-layer perceptron (MLP) from previous research [11], (2) a modified MLP with increased capacity and (3) a one-dimensional convolutional neural network (1D CNN). Hereafter, these models are referred to as MLP, mMLP, and CNN, respectively. Their frequency estimation accuracy and computational characteristics are compared using the same test dataset.

The remainder of this report is organized as follows. Section 2 introduces the signal model and previous research. Section 3 describes dataset generation and the proposed neural network architectures. Section 4 presents the numerical results and discusses model performance. Finally, Section 5 summarizes the findings and outlines future work.

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II. Background

A. Spacecraft Structural Vibrations and Signal Model

Spacecraft with flexible structures may exhibit vibration responses containing multiple frequency components. These components can be associated with structural modes, actuator disturbances, and other dynamic effects. In general, a vibration signal can be represented as a sum of sinusoidal components with potentially time-varying amplitudes and frequencies:

$$x(t) = \sum_{i=1}^{N} A_i(t) \sin\left(2\pi \int_0^t f_i(\tau)\, d\tau + \phi_i\right) + n(t) \tag{1}$$

where $A_i(t)$ is the amplitude, $f_i(t)$ is the instantaneous frequency, ϕ_i is the initial phase, and $n(t)$ is measurement noise.

Although actual spacecraft vibration signals may contain multiple components with time-varying amplitudes, this study focuses on a single sinusoidal component with a linearly increasing or decreasing frequency and additive Gaussian white noise. Only slowly time-varying frequencies are considered in this study, as changes in structural properties often lead to gradual variations in vibration frequencies. The simplified signal model is

$$x(t) = A \sin\left(2\pi \int_0^t f(\tau)\, d\tau + \phi_0\right) + n(t) \tag{2}$$

B. Previous Research: Constant-Frequency Estimation Using a Simple MLP

Previous research investigated neural network-based frequency estimation using noisy sinusoidal signals with constant frequencies. The network model consists of an input layer with 2,000 neurons (which is the number of signal's samples), three hidden layers with 2, 2, and 3 neurons, and a single output neuron for estimated frequency. The architecture is shown in Fig. 1. The simple network architecture demonstrated the feasibility of applying neural networks to frequency estimation using lightweight models and achieved a mean squared error of 4×10^{-5} for normalized frequencies in less than a second.

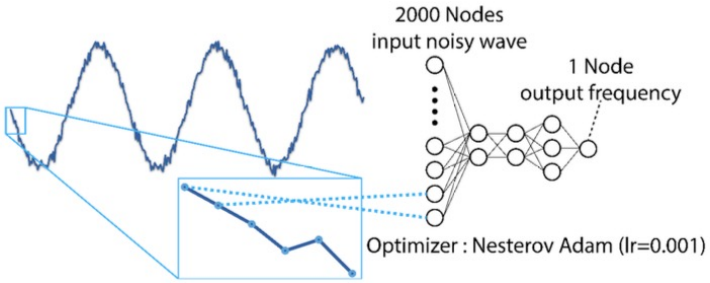


Fig. 1 Schematic of MLP model (image from [11]).

The original study focused on constant-frequency signals with a constant noise level (signal-to-noise ratio (SNR) of 25 dB). However, when the frequency varies over time, the output layer needs to be modified to estimate the changing frequencies. In addition, it is necessary to evaluate the robustness of the model under varying noise levels. Therefore, in this study, we aim to modify the architecture to estimate multiple intermediate frequencies, enabling the estimation of slowly time-varying frequencies.

III. Technical Approach

A. Dataset Generation

The training, validation, and test datasets are generated using MATLAB. Following the previous research paper [11], each signal contains 2,000 samples with a sampling interval of 1μs, corresponding to a sampling frequency of 1 MHz and a total observation duration of 2 ms. To reflect the linearly varying frequencies, a piecewise frequency profile is adopted, as shown in Eq.(3). The instantaneous frequency is initially constant, changes linearly during a transition

interval, and then remains constant at a final value.

$$f(t) = \begin{cases} f_1, & t < t_1 \\ f_1 + \dfrac{f_2 - f_1}{t_2 - t_1}(t - t_1), & t_1 \le t \le t_2 \\ f_2, & t > t_2. \end{cases} \tag{3}$$

The initial frequency f_1 is randomly selected between $1,000$ and $6,667$ Hz. The final frequency f_2 is determined by increasing or decreasing the initial frequency by a randomly selected percentage between 5% and 50%, resulting in a final frequency range between 500 and $10,000$ Hz. The transition duration is also varied between 30% and 90% of the total simulation time to introduce variations in the frequency change rate. Additive Gaussian noise is included to represent measurement uncertainty, with the SNR varued between 10 and 40 dB. The amplitude and initial phase are set to constant values of 1, and 0, respectively. Examples of a simulated signal is shown in Fig. 2.

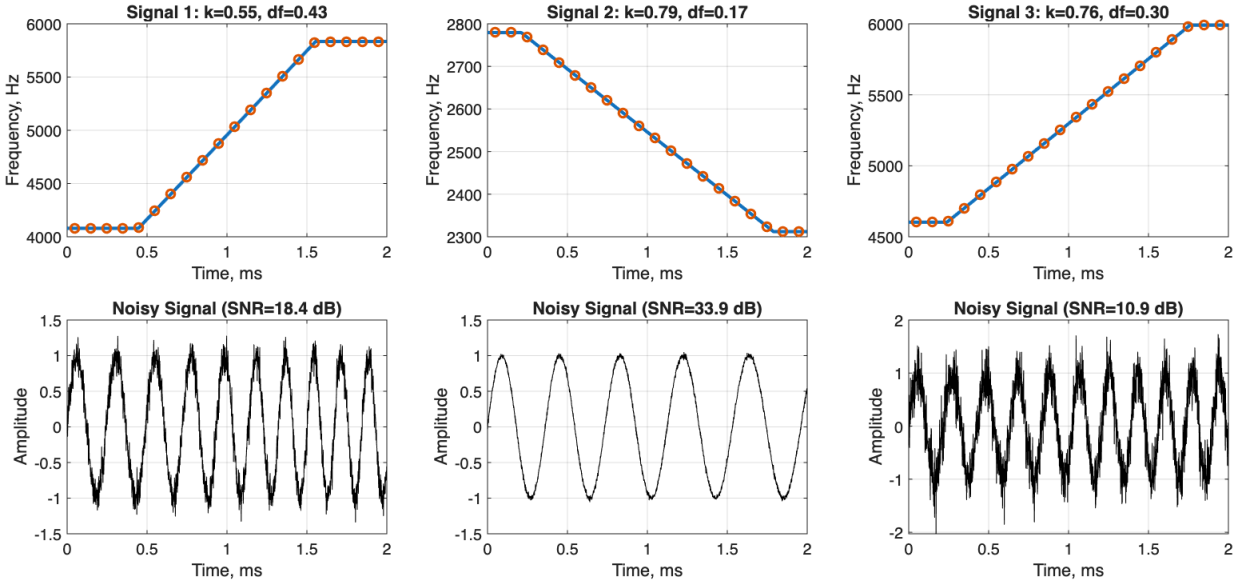
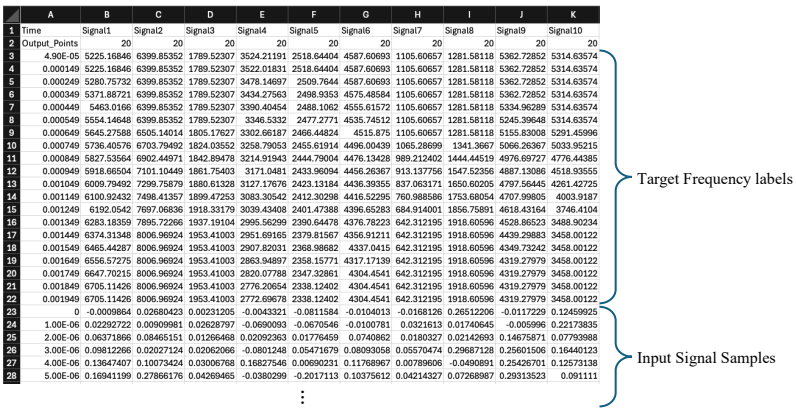


Fig. 2 Examples of generated signals with varying frequencies (k: transition ratio, df: frequency change ratio).

A total of $100,000$ synthetic signals are generated and divided into training, validation, and test datasets with ratios of 70%, 15%, and 15%, respectively. The target output consists of 20 instantaneous frequency values sampled at evenly spaced positions throughout the signal, which serve as the target values for the output layer. The target frequencies are normalized by a factor of $10,000$ during training to scale the frequency range to $[0, 1]$. The generated data are stored in CSV files containing signal samples, time values, and corresponding frequency labels. An example of a CSV file is reveled in Fig. 3.

B. Model 1: Original MLP

The first model is based on the simple MLP model from previous research. Its hidden-layer architecture was preserved as described in Section II.B, while the number of output neurons was changed from 1 to 20 to capture varying frequency profiles. The model was trained using the Nesterov-Adam optimizer (NAdam) with a learning rate of 0.001, and mean squared error (MSE) was used as the loss function. The parameters not specified in the original paper were set as follows: a batch size of 128 and the ReLU activation function for the hidden layers. To determine the optimal number of training epochs, the maximum number of epochs was set to 500, with early stopping applied if the validation loss did not improve for 30 consecutive epochs. The model with the lowest validation loss was saved for final evaluation. This model serves as a baseline to compare the performance of the following modified neural network architectures. Preliminary experiments indicate that the original architecture has difficulty learning the mapping from input signals to time-varying frequency profiles. This motivates the development of a modified MLP with increased capacity.



	A	B	C	D	E	F	G	H	I	J	K
1	Time	Signal1	Signal2	Signal3	Signal4	Signal5	Signal6	Signal7	Signal8	Signal9	Signal10
2	Output_Points	20	20	20	20	20	20	20	20	20	20
3	4.90E-05	5225.16846	6399.85352	1789.52307	3524.21191	2518.64404	4587.60693	1105.60657	1281.58118	5362.72852	5314.63574
4	0.000149	5225.16846	6399.85352	1789.52307	3522.01831	2518.64404	4587.60693	1105.60657	1281.58118	5362.72852	5314.63574
5	0.000249	5280.75732	6399.85352	1789.52307	3478.14697	2509.7644	4587.60693	1105.60657	1281.58118	5362.72852	5314.63574
6	0.000349	5371.88721	6399.85352	1789.52307	3434.27563	2498.9353	4575.48584	1105.60657	1281.58118	5362.72852	5314.63574
7	0.000449	5463.0166	6399.85352	1789.52307	3390.40454	2488.1062	4555.61572	1105.60657	1281.58118	5334.96289	5314.63574
8	0.000549	5554.14648	6399.85352	1789.52307	3346.5332	2477.2771	4535.74512	1105.60657	1281.58118	5245.39648	5314.63574
9	0.000649	5645.27588	6505.14014	1805.17627	3302.66187	2466.44824	4515.875	1105.60657	1281.58118	5155.83008	5291.45996
10	0.000749	5736.40576	6703.79492	1824.03552	3258.79053	2455.61914	4496.00439	1065.28699	1341.3667	5066.26367	5033.95215
11	0.000849	5827.53564	6902.44971	1842.89478	3214.91943	2444.79004	4476.13428	989.212402	1444.44519	4976.69727	4776.44385
12	0.000949	5918.66504	7101.10449	1861.75403	3171.0481	2433.96094	4456.26367	913.137756	1547.52356	4887.13086	4518.93555
13	0.001049	6009.79492	7299.75879	1880.61328	3127.17676	2423.13184	4436.39355	837.063171	1650.60205	4797.56445	4261.42725
14	0.001149	6100.92432	7498.41357	1899.47253	3083.30542	2412.30298	4416.52295	760.988586	1753.68054	4707.99805	4003.9187
15	0.001249	6192.0542	7697.06836	1918.33179	3039.43408	2401.47388	4396.65283	684.914001	1856.75891	4618.43164	3746.4104
16	0.001349	6283.18359	7895.72266	1937.19104	2995.56299	2390.64478	4376.78223	642.312195	1918.60596	4528.86523	3488.90234
17	0.001449	6374.31348	8006.96924	1953.41003	2951.69165	2379.81567	4356.91211	642.312195	1918.60596	4439.29883	3458.00122
18	0.001549	6465.44287	8006.96924	1953.41003	2907.82031	2368.98682	4337.0415	642.312195	1918.60596	4349.73242	3458.00122
19	0.001649	6556.57275	8006.96924	1953.41003	2863.94897	2358.15771	4317.17139	642.312195	1918.60596	4319.27979	3458.00122
20	0.001749	6647.70215	8006.96924	1953.41003	2820.07788	2347.32861	4304.4541	642.312195	1918.60596	4319.27979	3458.00122
21	0.001849	6705.11426	8006.96924	1953.41003	2776.20654	2338.12402	4304.4541	642.312195	1918.60596	4319.27979	3458.00122
22	0.001949	6705.11426	8006.96924	1953.41003	2772.69678	2338.12402	4304.4541	642.312195	1918.60596	4319.27979	3458.00122
23	0	-0.0009864	0.02680423	0.00231205	-0.0043321	-0.0811584	-0.0104013	-0.0168126	0.26512206	-0.0117229	0.12459925
24	1.00E-06	0.02292722	0.00909981	0.02628797	-0.0690093	-0.0670546	-0.0100781	0.0321613	0.01740645	-0.005996	0.22173835
25	2.00E-06	0.06371866	0.08465151	0.01266468	0.02092363	0.01776459	0.0740862	0.0180327	0.02142693	0.14675871	0.07793988
26	3.00E-06	0.09812266	0.02027124	0.02062066	-0.0801248	0.05471679	0.08093058	0.05570474	0.29687128	0.25601506	0.16440123
27	4.00E-06	0.13647407	0.10073424	0.03006768	0.16827546	0.00690231	0.11768967	0.00789606	-0.0490891	0.25426701	0.12573138
28	5.00E-06	0.16941199	0.27866176	0.04269465	-0.0380299	-0.2017113	0.10375612	0.04214327	0.07268987	0.29313523	0.091111

Fig. 3 Example of a generated CSV file output.

C. Model 2: Modified MLP

To address the limitation of the original architecture, a modified MLP is introduced. It contains three hidden layers with 128, 64, and 32 neurons, respectively. Other parameters, including the ReLU activation functions, are kept the same as those of the original MLP. The increased number of hidden neurons allows the model to learn more complex nonlinear relationships between input signal samples and instantaneous frequencies. The improved results are presented in Section IV. Although the modified framework improves estimation accuracy compared with the original architecture, it does not explicitly account for correlations between neighboring signal samples. Since the target frequency varies slowly over time, local temporal relationships may provide useful information for frequency estimation. Therefore, a network architecture that can explicitly capture these temporal correlation is considered.

D. Model 3: 1D CNN

Convolutional neural networks (CNN) are designed to extract local futures by applying convolutional filters across the input data. Since the input signals are one-dimensional time-series data, 1D CNN is adopted. This architecture is expected to better capture temporal features relevant to the estimation of slowly time-varying frequencies. The proposed CNN consists of three convolutional layers with 8, 24, and 8 output channels, respectively. All convolutionaly layers use a kernel size of 7 and padding of 3, followed by ReLU activation and max pooling with a factor of 2. The extracted features are than flattened and passed through a fully connected layer with 64 neurons, followed by an output layer with 20 neurons to estimate a time-varying frequency profile consisting of 20 points.

The architecture was selected through empirical evaluation of several CNN configurations, with the aim of achieving high estimation accuracy while maintaining a relatively small number of trainable parameters. Based on the observed estimation performance and model size, the configuration with 8-24-8 convolutional filters with kernel size of 7, and 64 fully connected neurons was selected. The resulting CNN contains approximately $132,000$ parameters, which is about half the number of parameters in the mMLP (approximately $267,000$ parameters). However, a lower parameter count does not necessarily translate into reduced computational time, as convolutional filters are repeatedly applied across the input signal, increasing the computational workload. The training settings, including the optimizer, learning rate, batch size, loss function, and early stopping criteria, are kept the same as those used for the MLP models. Fig. 4 shows the 1D CNN full architecture for slowly time-varying frequency estimation.

E. Training and Evaluation Procedure

All three models are trained using the same synthetic training and validation datasets. The mean squared error (MSE) between predicted and true normalized frequencies is used for performance evaluation. Training is conducted for up to 500 epochs, with early stopping applied when the validation loss does not improve. The best model parameters are saved based on validation performance. The final models are evaluated using $15,000$ independent test signals. Estimation accuracy is assessed using mean absolute error (MAE) and root mean squared error (RMSE). In addition to estimation accuracy, the total training and validation time and test time are measured to compare the computational efficiency of the three models. All experiments are conducted on a MacBook Pro with an Apple M5 chip, using GPU acceleration via MPS.

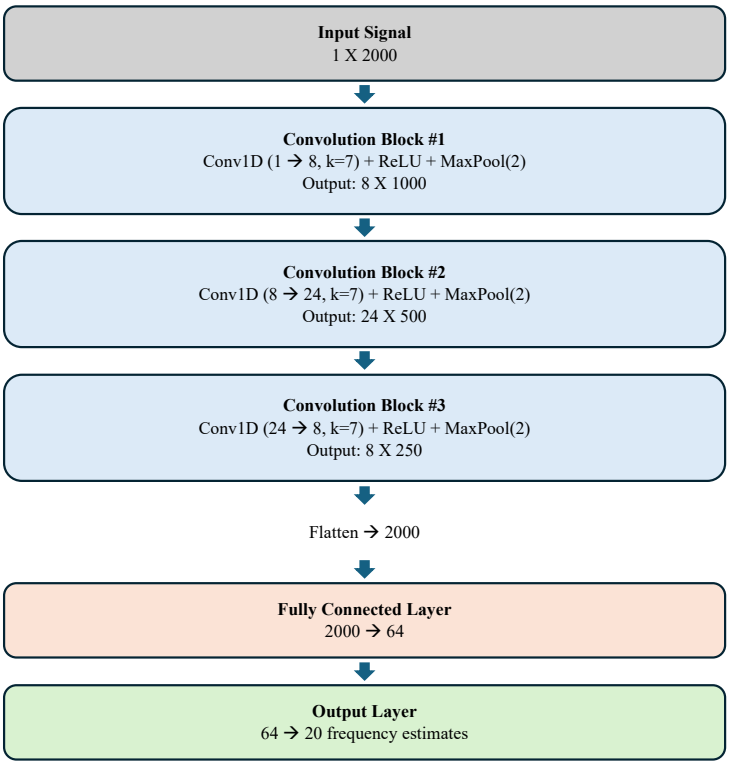


Fig. 4 Architecture of the proposed 1D CNN.

IV. Results

A. Training and Validation Results

The three models are trained using the training and validation datasets. Before evaluating their performance on time-varying frequency signals, a preliminary comparison was conducted using constant-frequency signals, as considered in the previous study [11]. The input signals contained randomly selected frequencies ranging from $1,000$ to $10,000$ Hz, with randomly assigned SNRs between 10 and 40 dB. The datasets were generated using the same data selection procedure described in Section III.A. For this single-frequency estimaiton task, the output layer of each model was modified to contain a single neuron, while the remaining architecture was kept unchanged. The validation RMSE values were 46.581, 5.842, and 10.392 Hz for MLP, mMLP, and CNN, respectively. The result obtained with the MLP differed from that reported in the previous study; this difference may be attributed to differences in hyperparameter settings and noise conditions. Therefore, these results serve as a baseline comparison of the three architectures for single-frequency estimation.

Table 1 shows the training and validation for time-varying frequency estimation, along with the computation times. The total time represents the entire training duration required to find the best epoch and obtain the best model, while the average time represents the computation time per epoch.

The original MLP cannot track the time-varying frequency profiles, including the initial frequencies. This may be

Table 1 Training and validation performance comparison.

Model	Best Epoch	Validation MAE (Hz)	Validation RMSE (Hz)	Total Time (s)	Avg. Time (s)
Original MLP	7	1521.818	1816.601	29.47	0.7964
Modified MLP	158	14.345	20.457	167.80	0.8926
1D CNN	249	**10.417**	**14.986**	1384.80	4.9635

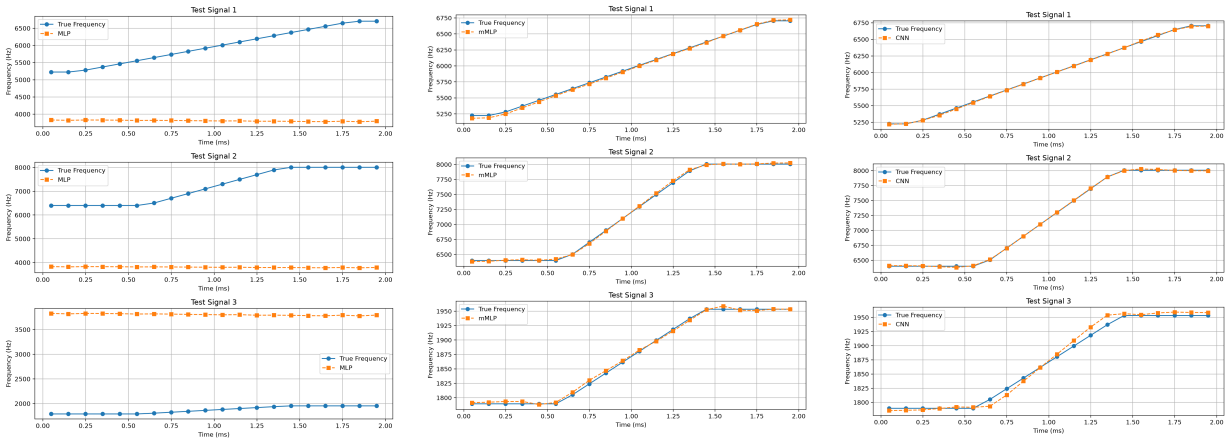
attributed to its limited model capacity, which restricts its ability to learn the relationship between input signals and their corresponding frequency profiles. Similar results are observed even when early stopping is disabled. In contrast, the mMLP significantly reduces the validation errors compared with the original MLP, while maintaining a relatively low computation time per epoch. The 1D CNN achieves the lowest validation MAE and RMSE among the three models, although it requires a substantially longer training time.

B. Frequency Estimation Performance

The three models are evaluated on the same test dataset containing 15,000 signals. Each model predicts 20 instantaneous frequency values per signal, resulting in a total of 300,000 frequency estimates. The test results are summarized in Table 2, and examples of the frequency estimation results are shown in Fig. 5.

Table 2 Performance comparison of neural network models for slowly time-varying frequency estimation.

Model	Test MAE (Hz)	Test RMSE (Hz)	Total Test Time (s)
Original MLP	1509.994	1808.618	0.0866
Modified MLP	14.531	20.823	0.0861
1D CNN	**10.351**	**14.845**	0.431



(a) (b) (c)

Fig. 5 Frequency estimation results of the three models. (a) MLP, (b) mMLP, and (c) CNN.

The original MLP exhibits poor performance on time-varying frequency signals. Its predictions remain nearly constant or vary only slightly, failing to adequately track the true frequency profiles, consistent with the validation results. In contrast, the mMLP substantially improves the estimation accuracy, achieving a test MAE of 14.531 Hz and an RMSE of 20.823 Hz. The 1D CNN further reduces these errors to 10.351 Hz and 14.845 Hz, respectively, achieving the best estimation performance among the three models. Given the frequency range of 1,000 - 10,000 Hz, these errors correspond to approximately 0.103% and 0.148% of the maximum frequency, respectively. Compared with the mMLP, the CNN reduces both the test MAE and RMSE by approximately 29%.

The test errors of all three models are close to their corresponding validation errors, suggesting that the trained models generalize consistently to unseen signals generated under the same synthetic data conditions. In terms of computational efficiency, the MLP and mMLP require similar inference times per signal, whereas the CNN requires approximately five times longer. These results demonstrate a trade-off between estimation accuracy and computational cost, with the CNN providing the highest accuracy and the mMLP offering lower computational cost with slightly higher estimation errors.

As shown in Fig. 5, both the mMLP and CNN closely follow the true frequency profiles in the first two examples, with the CNN providing more accurate estimates. However, in the third example, which has a relatively small frequency

variation (df = 0.09), the mMLP provides more accurate estimates than the CNN. Although the CNN achieves the lowest overall MAE and RMSE, these metrics are averaged over all 300, 000 frequency estimates and do not guarantee superior performance for every individual signal. This suggests that the relative performance of the models may depend on the characteristics of the frequency variation.

C. Preliminary Evaluation Using Real Bearing Vibration Data

A preliminary investigation was conducted using the bearing vibration dataset [12] to explore the applicability of the trained neural network models to real vibration measurements. The dataset contains vibration signals collected from a healthy bearing under increasing (H-A-1, H-A-2, and H-A-3) and decreasing (H-B-1, H-B-2, and H-B-3) rotational speeds ranging from 11.6 to 29.0 Hz, with sampling rate at 200 kHz. To make the real data more compatible with the synthetic training signals, the sampling rate was adjusted so that the frequencies corresponded to the 1, 000 - 10, 000 Hz frequency range of the training data. Subsequently, 2, 000 data points were extracted from each signal to match the input size of the trained neural network models. Unlike the synthetic training signals, these experimental measurements contain complex vibration components and noise, providing a more challenging condition for frequency estimation, as shown in Fig. 6.

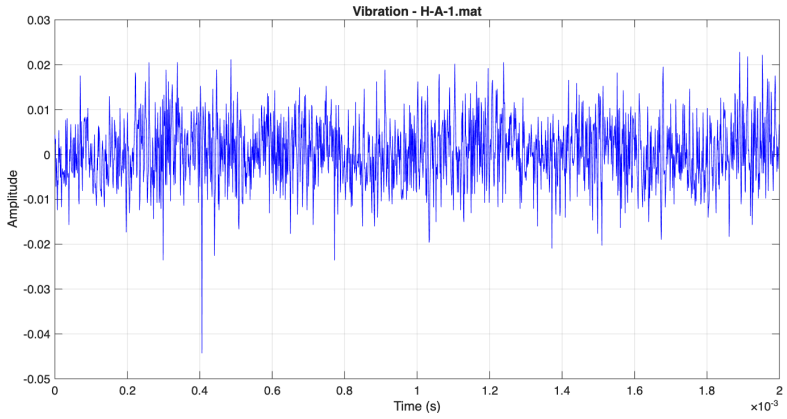
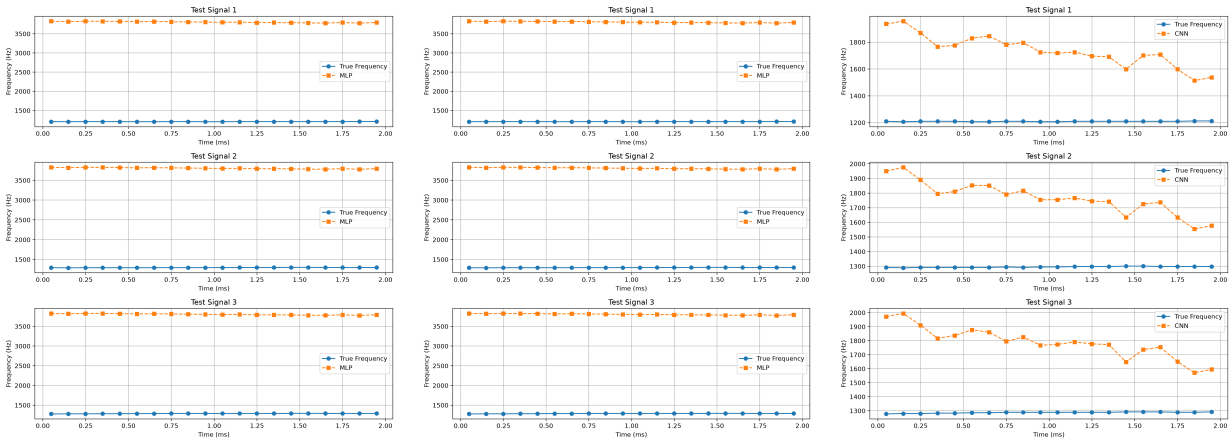


Fig. 6 Example of real bearing vibration signal.



(a) (b) (c)

Fig. 7 Frequency estimation results of the three models. (a) MLP, (b) mMLP, and (c) CNN.

As shown in Fig. 7, the preliminary results showed that all three models experienced a significant decrease in estimation accuracy when applied to real bearing vibration data. The MLP and mMLP achieved RMSE values of

2560.95 Hz and 2253.68 Hz, respectively, while the CNN achieved a substantially lower RMSE of 555.76 Hz. Although the CNN got the best results, its performance was considerably worse than on the synthetic test dataset. This degradation suggests that the models trained on simplified sinusoidal signals have limited generalization capability when applied to complex real-world vibration measurements.

V. Conclusions

This study investigated neural network-based estimation of time-varying frequencies. Three architectures were evaluated: the original MLP adapted from previous constant-frequency research, a modified MLP, and a 1D CNN. The original MLP showed poor performance on time-varying frequency signals, suggesting that its limited network capacity was insufficient to capture the more complex frequency variations. Increasing the network capacity substantially improved the estimation accuracy, reducing the test RMSE to 20.823 Hz. The 1D CNN achieved the best performance, with a test RMSE of 14.845 Hz, representing a 29% improvement compared with the modified MLP.

These results suggest that convolutional feature extraction is advantageous for time-varying frequency estimation, Unlike fully connected networks, CNNs can exploit local temporal patterns through shared convolutional filters, allowing frequency-related features to be extracted efficiently in terms of model parameters. However, the CNN required significantly more computational time than the mMLP structure, indicating a trade-off between estimation accuracy and computational cost.

Although the 1D CNN demonstrated high accuracy on synthetic signals, its performance deteriorated when applied to real experimental signals. This limitation highlights the gap between simplified signals and real vibration measurements, which may contain multiple frequency components, structural resonances, and complex noise. Therefore, the current results demonstrate the feasibility of the approach under controlled synthetic conditions rather than its readiness for real-world implementation.

The main contribution of this study is to demonstrate that extending neural network-based frequency estimation from constant-frequency signals to time-varying frequency signals requires sufficient network capacity, and that a 1D CNN can improve estimation accuracy with fewer trainable parameters. However, a key limitation of this study is the assumption that vibration signals consist of a single time-varying frequency component with simple additive noise, which does not fully represent complex real-world vibrations. Future work will focus on extending the approach to multi-frequency signals with time-varying amplitudes and frequencies and incorporating more realistic noise characteristics into the training data. In addition, network architectures and training strategies will be investigated to improve estimation accuracy and robustness when applied to real experimental measurements. These developments are expected to support more robust frequency estimation in various applications, including spacecraft structural vibration monitoring.

Acknowledgments

ChatGPT was used to assist in writing MATLAB and Python scripts for this study. All scripts were developed based on the author's instructions and were subsequently reviewed and verified by the author. The author takes full responsibility for the implementation and results of the code.

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